

Educational Insights with PISA 2022 Datasets

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Abstract

Education has been and remains one of the most fundamentally important services provided by governments today. Historical analysis has drawn strong connections between the quality of an individual's education and the quality of their later-life outcomes. However, education not only affects an individual's outcome but also national level competitiveness. By examining the factors that influence and predict student performance on mathematics, reading, and science proficiency countries can focus their efforts and policy initiatives to influence the areas that will have the greatest impact on student success.

This study uses the comprehensive PISA 2022 dataset to investigate the predictors of student academic performance, with a focus on math proficiency at the country level. We examined three aims: (1) whether average reading proficiency and behavioral factors predict average math performance; (2) identifying key predictors of average minimum math proficiency; and (3) exploring the relationship between average national math proficiency levels and the prevalence of food insecurity.

To address these aims we applied 5 different models: linear regression for Aim 1; Gaussian Naive Bayes and Support Vector Machine (SVM) for Aim 2; and Random Forest and K-nearest neighbors (KNN) for Aim 3. Hyperparameter tuning was applied where appropriate and the resulting optimized models were compared using applicable metrics. Linear regression models were compared using AIC, BIC, RMSE, and

adjusted R^2 . Our other models were compared using precision, recall, and F-1 scores as well as Receiver Operating Characteristic - Area Under the Curve (ROC-AUC).

For Aim 1, our findings indicate that for Linear Regression reading proficiency is the strongest predictor of math performance. The subjective love of learning of a student improved the predictive nature of the model, but the effect was minimal. For Aim 2, Gaussian Naive Bayes outperformed SVM, with average socioeconomic status and food insecurity emerging as key predictors. For Aim 3, Random Forest outperformed KNN in classifying countries by three categories; "failed to meet minimum proficiency", "meet minimum proficiency", or "exceed minimum proficiency" based on food insecurity. It is worth noting that both models struggled to accurately predict countries in the "met minimum proficiency" category better than chance. Interestingly, food insecurity was ranked behind socioeconomic status and national GDP in predictive importance.

These results highlight the influence of socioeconomic and systemic factors on academic performance and offer data-driven insights to guide educational policy. However, due to constraints within the datasets, these results represent country level insights. Future work should explore these relationships at the more granular student level. The results have the potential to further support our findings as well as uncover new connections that could better inform targeted interventions.

1 Introduction and Related Work

Our research aims to draw conclusions and insights from the Programme for International Student Assessment (PISA), a survey developed and administered by the Organisation for Economic Co-operation and Development (OECD) every 3 years. This survey tests the skills of 15-year old students in the areas of mathematics, reading, and science. In total, eighty-one countries participated in the 2022 survey, which had a more specific focus on the area of mathematics.

PISA provides an unbiased evaluation of student performance in key educational areas, as well as an assessment of their habits, life skills and mental health. These measures can be used to interpret the performance of the country's overall educational system and educational policies. Additionally, teenagers are a country's future and education serves as the foundation for skills required within the modern workforce. In 15 to 20 years, that cohort will be that country's innovators, educators, entrepreneurs, and rising CEOs. With globalized economic systems, a country's future workforce can be a strong indicator of that country's ability to compete on a global scale.

Education has been and remains one of the most fundamentally important services provided by governments. Historical analysis has drawn strong connections between the quality of an individual's education and the quality of their later-life outcomes. Countries can use this information to see where their students stack up on a global scale and can use their results to honestly evaluate their educational systems and processes, gaining insights into where improvements can be made. Results may also drive international cooperation on educational reform and improvement, allowing lower performing countries for a given measure to learn from their higher performing peers, emulating their approaches or techniques for educational policy. Additionally, this version of the PISA focused specifically on mathematics achievements. As such, university programs and industries which require high levels of STEM proficiency would be particularly interested in these results. Patterns that indicate relationships between educational outcomes and factors not directly tied to the school room could lead to a greater integration between the educational process and other facets of daily life.

The PISA 2022 Results (Volume V) outlines five key solutions to improving academic performance. Firstly, socio-economically disadvantaged students struggle to use learning strategies that foster success. Schools should implement programs to provide ad-

ditional support to these students, such as tutoring and parental workshops. Equally important, bridging the digital divide provides access to online information and builds digital literacy. In addition to the educational benefits of online access, digital literacy helps students differentiate truth from fiction and act responsibly online. The report also recommends tailoring education to varying types of learners. Schools should assess individual student needs and develop strategies to support a variety of learning styles to build confidence, resilience and adaptability among students.

The PISA has also consistently shown gender disparities, with boys outperforming girls in mathematics and girls surpassing boys in self-monitoring strategies. Girls also exhibit higher levels of anxiety. One solution to this ongoing problem is reducing gender-based stereotypes and encouraging equal opportunities for boys and girls.

Finally, student engagement is directly linked to strong teacher-student relationships. Teacher support helps students grow in independence and critical thinking. Students also feel more connected to their education and experience less anxiety, which improves academic outcomes.

Although we know what works, there are several challenges in implementing these solutions including consistent and adequate funding, supporting our teachers, mental health challenges among students, and appropriate use of technology. Many communities across the globe lack reliable funding to provide high-quality education to students and retain teachers. Social media and technology pose additional challenges to students, who are already in a vulnerable life stage. Teens experience high levels of anxiety and depression, which negatively impacts their ability to succeed in school.

Initial exploratory data analysis (EDA) focused on uncovering relationships between features of the dataset guided by 10 preliminary research questions, as follows:

1. What is the relationship between gross domestic product (GDP) and socioeconomic (SES) disparities with life satisfaction as a whole among students?
2. What factors influence countries meeting the minimum proficiency level in reading, mathematics and science (Sustainable Development Goal 4.1.1¹⁰)?
3. How does a growth mindset relate to math, reading and science proficiency?

4. How does grit, love of learning, and empathy among students influence Science proficiency?
5. What effect does daily moderate to vigorous exercise have on the mental health of students?
6. How does total government expenditure on education influence reading and math proficiency?
7. How does food availability affect student performance generally?
8. How does literacy level impact student performance in mathematics?
9. How much of the variation in student performance is explained by socioeconomic status of the individual? Of the country of origin?
10. What impact does socioeconomic status have on school attendance, and which has the greater independent impact on academic performance?

Upon EDA completion, we concluded that the data supported analysis of academic performance, as measured by math proficiency, and our ability to predict these scores based on a number of features. We pursued three aims, related to questions 2, 7 and 8 above, as follows:

- Aim 1: To determine whether the average reading score of a country, and several other behavioral characteristics, are predictive of average math proficiency.
- Aim 2: To determine the best predictors of average minimum math proficiency.
- Aim 3: To determine the relationship between a country's average math proficiency level, categorized as below minimum proficiency, meeting minimum proficiency and above minimum proficiency, with the average proportion of food insecurity.

2 Methods, Evaluation, and Results

2.1 Data Sources, Cleaning, Preprocessing, and Feature Selection

In 2022, over half a million students from 81 countries completed the PISA. We rolled-up student-level responses to combine these data with country-level PISA scores, GDP, and percent GDP spent on education (Figure 1).

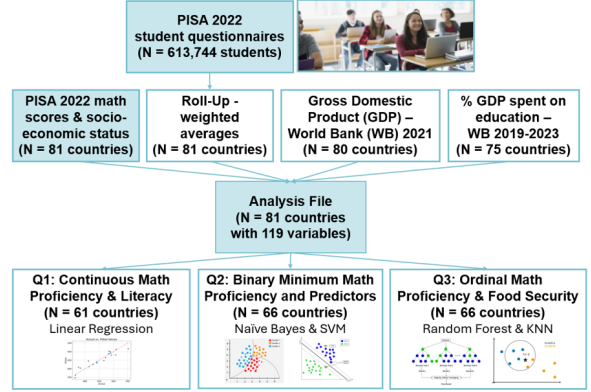


Figure 1: Data Sources and Analysis Datafile Attributes

Data were collected from multiple sources, as follows:

- PISA student testing and SES data were collected from PISA 2022 Results (Volume I) OECD 2023, Appendix B1. PDF tables were converted to Excel files⁶
- PISA student questionnaire data were downloaded from PISA after completing a data use agreement⁷
- GDP by country came from the World Bank and was accessed through Our World in Data⁸
- The percent GDP spent on education by country also came from the World Bank and was available through UNESCO⁹

As we collected and combined the data, we conducted data cleaning and preprocessing to ensure that our data would meet the necessary standards to allow for us to answer the questions that we posed at the beginning of our investigation. We addressed missing values by replacing them with the median when they were rare, and excluding the countries with missing data when common. For example, food security was missing for 15 countries because the country did not include these questions in their student questionnaire. These countries were removed from analyses which included food security.

Our primary questions revolve around how various factors influence student performance as measured by the 2022 PISA assessment. For these academic proficiency outcomes we analyzed mean math, reading and science scores by country. From our exploratory data analysis, we see that the average math scores range from 336 to 575 with a mean(sd) of 437.65(58.19) and median(P25, P75) of 441(389,

483). We created z-scores for these proficiency measures and reviewed these QQ-plots. Figure 2 provides the QQ-Plot for the normalized average math scores, which did not appear normal (Shapiro-Wilk test for normality, $p=0.027$). We also normalized continuous SES and GDP measures.

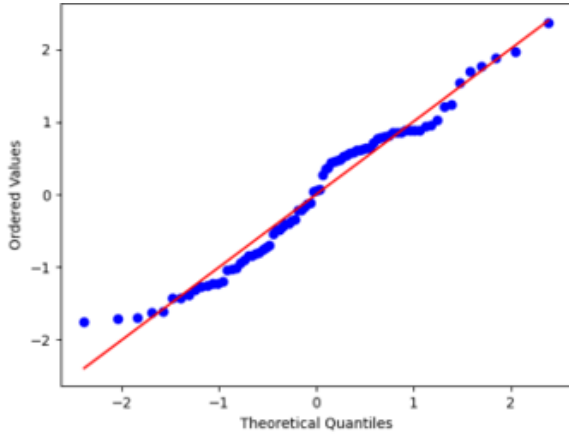


Figure 2: QQ-Plot of Normalized Average PISA Math Proficiency Scores

In regards to dimensionality reduction, our main methods of reduction were through rolling-up the student data ($N=613,744$) to the country level ($N=81$) using weighted averages, combining variables into meaningful proportions, and using variable selection. Weighted proportions were calculated using the trimmed, final, non-response adjusted student weights. We calculated summary statistics (mean, median, variance, skewness, etc.), correlations between features, and completed data similarity and integration analyses. For feature selection, we chose some variables based on their relationship with math proficiency. We found that the average math, reading and science scores are highly positively correlated with each other $r > 0.9$. GDP and SES are also strongly positively correlated with these scores ($r = 0.7$), while lack of grit (ST307 - students strongly disagreed with, “I apply additional effort when work becomes challenging”) and lack of empathy (ST303 - students strongly disagreed with, “I imagine how I would feel if I were in somebody else’s place”) are negatively correlated with the achievement scores. Never or almost never going hungry (ST258) was moderately positively correlated with these scores $r \sim 0.6$.

For each of our study aims, we completed EDA and chose the most appropriate model type(s) to further explore the relationships. Specifically, we completed the following:

- Aim 1: continuous math score and literacy (lin-

ear regression)

- Aim 2: binary minimum math proficiency and predictive features through Gaussian Naïve Bayesian and Support Vector Machines
- Aim 3: math proficiency level with food insecurity through Random Forest and K-nearest neighbors

For each model we chose the best features, tuned hyper-parameters, and evaluated metrics of model fit and performance. For each aim we determined the best model and interpreted the findings.

2.2 Aim 1 Evaluation and Results: Relationship between Literacy and Math Scores

Our first aim was to assess the degree to which a student’s literacy level can predict their performance in mathematics. In addition to this, we also included several other behavioral variables that were available through the PISA data to see if their addition to the model could better help predict math performance. To explore this relationship, we collated country-level information on average reading proficiency and math proficiency. We then plotted these against each other by country. From Figure 1 below, it is clear that there is a strong, positive, linear relationship between these variables.

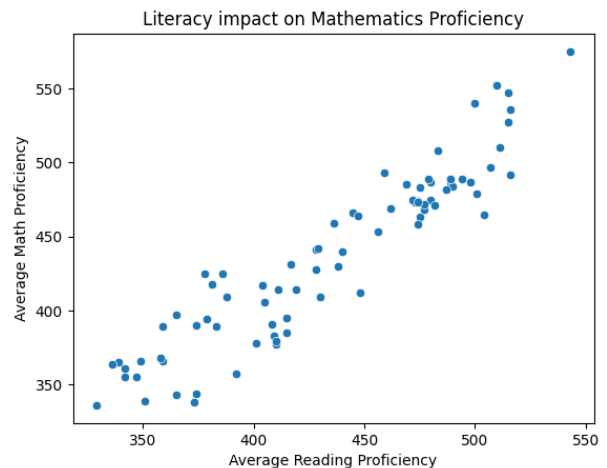


Figure 3: Relationship Between Literacy and Math Proficiency

Based on the perceived relationship in 1, we concluded that a linear regression model would be the most apt model to predict math proficiency scores. A potential strong positive, linear relationship lends

well to this type of model. Additionally, with this model we would also be able to incorporate other variables of interest to see whether they have any predictive power on math performance.

In using this model, we made several assumptions. The first assumption was that a linear relationship between the dependent variable, math performance, and the independent variables, reading performance, socioeconomic status score, a hunger variable, a growth mindset variable, a love of learning variable, and percent of GDP spent on education, was a valid structure for our model. The second assumption was that the chosen behavioral variables actually had some impact on a student’s math performance score.

Additionally, prior to the construction of the models we assumed that the linear regression models would conform to the required assumptions of linearity, homoscedasticity, and normality of residuals. Once the models were built, these assumptions were tested to ensure that the models were sufficient. The models passed these three tests, confirming that a linear regression model was an apt model for prediction.

The main adjustment made during model development to improve performance was adjusting the number of variables being considered in model construction. We started with just considering a model that only included Reading Performance as an independent variable. The summary of this model can be seen in Table 1 below. As expected, the model performed well and supported our initial conclusion that literacy level would be a strong, positive predictor of math performance. The model had an R^2 value of 0.848.

OLS Regression Results						
Dep. Variable:	math_all_mean	R-squared:	0.848			
Model:	OLS	Adj. R-squared:	0.839			
Method:	Least Squares	F-statistic:	95.04			
Date:	Sun, 13 Apr 2025	Prob (F-statistic):	2.24e-08			
Time:	21:53:34	Log-likelihood:	-84.358			
No. Observations:	19	AIC:	172.7			
Df Residuals:	17	BIC:	174.6			
Df Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	-3.7625	46.939	-0.080	0.937	-102.795	95.270
read_all_mean	1.0196	0.105	9.749	0.000	0.799	1.240
Omnibus:	0.546	Durbin-Watson:	1.754			
Prob(Omnibus):	0.761	Jarque-Bera (JB):	0.540			
Skew:	-0.338	Prob(JB):	0.764			
Kurtosis:	2.527	Cond. No.	4.23e+03			

Table 1: Simple Linear Regression for Average Math Proficiency by Average Reading Proficiency

Next, we constructed a model that included all of the additional independent variables mentioned above. The format of these independent variables were such that each independent variable was asso-

ciated with 5 or 8 sub-variables. For instance, for love of learning a student had the ability to choose between 5 different answers ranging from “strongly disagree” to “strongly agree”. Each sub-variable represented the percentage of students in that particular country that chose that answer for that question. Therefore, if the values for each sub-variable were summed, the result would be 1, accounting for 100 percent of the students in that country. This resulted in 17 total independent variables. Due to the size of our dataset and the resulting size of our training and testing sets, this model only had 1 residual degree of freedom. This caused the model to be extremely overfit. Therefore, we were not able to utilize this model due to our dataset constraints.

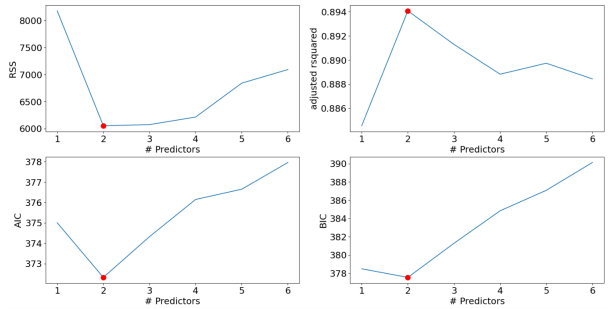


Figure 4: Model Selection for the Best Number of Predictors using RSS, Adjusted R^2 , AIC, and BIC

In light of this constraint, we constructed aggregate variables for each behavioral variable to reduce the number of parameters in the model and increase the number of residual degrees of freedom. This yielded a model structure that we could analyze and further improve. The aggregate variable for hunger was the percentage of students who reported not eating 4 or more times a week in the last 30 days due to not having enough money. The aggregate variable for growth mindset was the percentage of students that have a high growth mindset, which means answering agree or strongly agree to the relevant question. The aggregate variable for love of learning was the percentage of students that have a high love of learning, which means answering agree or strongly agree to the relevant question. Once aggregation was complete, we conducted subset regression analysis to determine the best model by considering every possible combination of variables from the 7 total variables. The comparison of these models can be seen in Figure 4. This comparison showed that the two variable model, which included the reading performance variable and the Love of Learning aggregate variable, was the superior model when considering the reduction of RSS, AIC, BIC, and the maximization of adjusted R^2 . Ta-

ble 2 displays the summary output from this optimal model.

OLS Regression Results						
Dep. Variable:	math_all_mean	R-squared:	0.899			
Model:	OLS	Adj. R-squared:	0.894			
Method:	Least Squares	F-statistic:	174.0			
Date:	Sun, 13 Apr 2025	Prob (F-statistic):	3.66e-20			
Time:	21:53:38	Log-Likelihood:	-183.16			
No. Observations:	42	AIC:	372.3			
DF Residuals:	39	BIC:	377.5			
DF Model:	2					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	144.4678	59.695	2.420	0.020	23.723	265.212
read_all_mean	0.8530	0.070	12.228	0.000	0.712	0.994
Percent High Love of Learning	-102.1263	47.631	-2.144	0.038	-198.468	-5.784
Omnibus:	1.797	Durbin-Watson:				
Prob(Omnibus):	0.407	Jarque-Bera (JB):				
skew:	0.454	Prob(JB):				
Kurtosis:	2.871	Cond. No.				

Table 2: Multiple Linear Regression for Average Math Proficiency by Average Reading Proficiency and High Love of Learning

We had now narrowed down to two models, our original model that only used reading performance as a parameter and our optimized two parameter model above. We used adjusted R^2 , AIC, BIC, and RMSE to evaluate and compare the models, the results of which can be seen in Table 3. Based on the resulting metrics, we concluded that the two parameter model with the reading performance variable and the aggregate love of learning variable was the best model. This was based on a better adjusted R^2 value as well as a better RMSE. AIC and BIC were higher for our chosen model compared to the other model. However, we know that both AIC and BIC penalize a model for its complexity, yielding a higher value. Since our chosen model only had 2 parameters compared to the other model's one parameter, we felt as though this additional complexity was not excessive and therefore discounted the worse performance in AIC and BIC measures. An actual vs. fitted plot for our chosen model can be seen in Figure 5.

	Model	Adj R ²	AIC	BIC	RMSE
0	Just Reading	0.848269	172.716448	174.605326	20.747491
1	Best Model w/ Aggregation Variables	0.894075	372.323663	377.536672	18.954805

Table 3: Model Selection

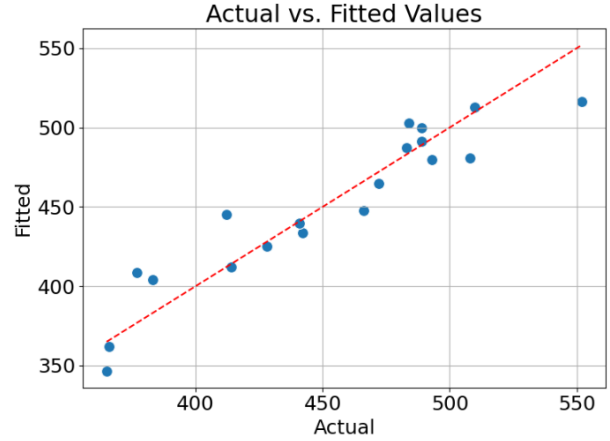


Figure 5: Actual Versus Fitted Math Proficiency for the Best Model

The main challenge that we faced during aim 1 was the size of our dataset. Originally, our dataset included 81 countries to examine. However, after cleaning the data and removing countries that were missing values for our variables of interest, we were left with 61 countries. We concluded this was an adequate size for model development, even though it presented problems during the development of our second model with 17 parameters. In the end, we concluded that our dataset was adequate and our investigation was still robust even without the ability to fully investigate a model with more parameters.

2.3 Aim 2 Evaluation and Results: Relationship between Minimum Math Proficiency and Food Availability, SES, Geographic Region, and Percent of GDP spent of Education

For our second question, we are interested in factors influencing countries meeting the minimum proficiency levels for academic success. This is one of the Sustainable Development Goals, specifically 4.1.1¹⁰. Minimum achievement is reaching at least a Level 2 on the exam. For math, this equates to earning a score of at least 420.07. We dichotomized the countries into those with an average math score of 420.07 or above, and those below this cutoff. We looked at economic factors by whether the countries met this minimum standard (Figure 6). For these binary outcomes, we selected Naive Bayesian classification and Support Vector Machines in order to explore the relationships between our features and educational outcomes.

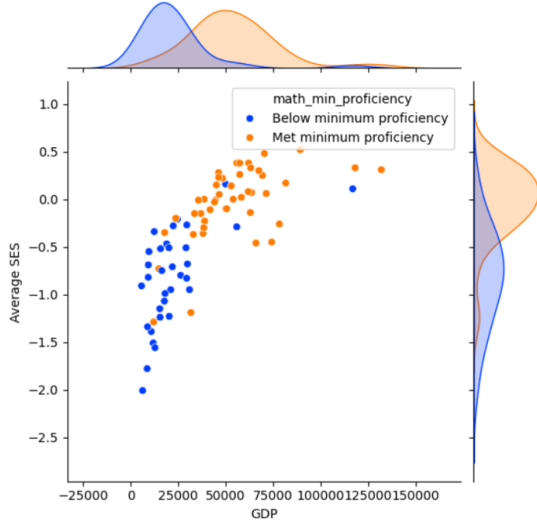


Figure 6: Minimum proficiency in math by country economic factors

We found that the World Bank GDP and the PISA student response SES variables corresponded well. With a few exceptions, those countries performing on average below the minimum proficiency cut off were also the poorest countries.

Naive Bayes classifiers in general assume conditional independence between features; that is, knowing the value of one feature does not provide any additional information about the value of other features. In addition, the Gaussian version of the model (which allows for the inclusion of continuous data as predictors) assumes that the continuous features of the model are normally distributed.

Support Vector Machine classification, on the other hand, relies on a few key assumptions. First, and most importantly, SVM assumes that the data being used is linearly separable; that is, that it can be separated into distinct classes by a hyperplane. SVM also assumes that model features are relatively independent.

All continuous features were normalized via StandardScaler. Prior to scaling, the summary of values for the variables of interest were as follows:

	ses_all_mean	ed_percent_gdp
count	66.000000	66.000000
mean	-0.439242	4.389062
std	0.594977	1.128043
min	-2.010000	1.275171
25%	-0.827500	3.593097
50%	-0.345000	4.615470
75%	0.042500	5.201258
max	0.480000	7.139665

Figure 7: Original Data Summary

After the use of StandardScaler (which produces a z-score), the scaled values were as follows:

	ses_all_mean	ed_percent_gdp
count	6.600000e+01	6.600000e+01
mean	1.328903e-16	-2.052230e-16
std	1.007663e+00	1.007663e+00
min	-2.660260e+00	-2.781590e+00
25%	-6.575593e-01	-7.110229e-01
50%	1.596105e-01	2.022470e-01
75%	8.158867e-01	7.255220e-01
max	1.556844e+00	2.457071e+00

Figure 8: Transformed Data Summary

Similar transformations were performed on the other continuous variables, with comparable results.

Both models were tuned using the GridSearchCV() tool from the SciKit-Learn library, which allows the user to provide a selection of different values for the model parameters, each of which is tested against the others to determine the optimum value from those tested.

For Gaussian Naive Bayes classifiers, there is only one hyperparameter to be tuned, called "var smoothing". The purpose of this parameter is to add a very small value to the variance of each continuous variable in order to avoid the potential problem of dividing by zero during calculations. The default value is 1e-9, and this value as well as others incremented up

by an order of magnitude through $1e-4$ were tested; however, the default value proved best for the model.

For Support Vector Machine, three hyperparameters were considered during tuning. The first, “C”, is also known as the regularization parameter, and controls how “hard” or “soft” the margin is; a softer margin is more tolerant to misclassification and corresponds to a smaller value of C, while a harder margin is less tolerant and corresponds to a higher value for C. During tuning we tested values of C ranging from 0.1 to 100, increasing by an order of magnitude at each step. The optimal value was found to be at 0.1. The second hyperparameter, “kernel”, refers to the specific type of hyperplane being generated to separate classes. The classic form (and the version explored in class) is a linear hyperplane, but others tested during tuning included “poly” (for polynomial) and “rbf” (for radial basis function). Of these three, rbf was found to be the best for the particular data set used. Finally, the third hyperparameter considered was “gamma”, which is a coefficient used with the rbf and poly kernels. This parameter controls the extent a single data point can influence the placement of the hyperplane. For this parameter, values of “scale” (which automatically determines the coefficient based on the number of features and the variance of the data), 0.01, 0.001, and 0.0001 were tested, with “scale” being selected as best for the data.

Model	Actual	Predicted: < min prof	Predicted: min prof	precision	recall	f1-score
Gaussian NB	< min prof	5	1	0.714286	0.833333	0.769231
Gaussian NB	min prof	2	6	0.857143	0.750000	0.800000
SVM	< min prof	0	6	0.000000	0.000000	0.000000
SVM	min prof	0	8	0.571429	1.000000	0.727273

Table 4: Evaluation metrics: Accuracy, Precision, Recall, F1-score, ROC-AUC

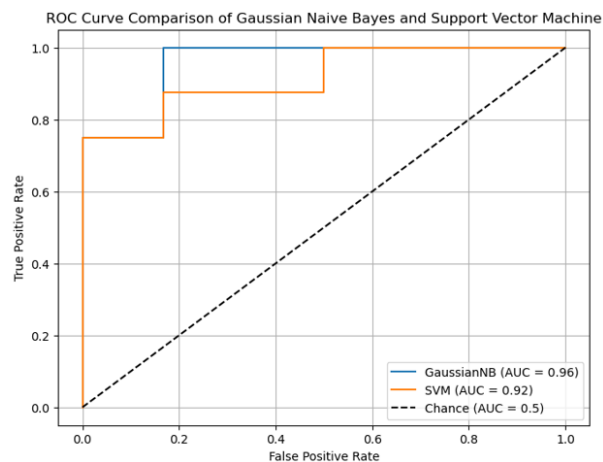


Figure 9: ROC Curve Comparison of Gaussian Naive Bayes and Support Vector Machine for Models of Minimum Math Proficiency

Based on our evaluation metrics, the Gaussian Naive Bayes model performed better for this data set on almost all metrics; unfortunately, even with tuning SVM wildly overpredicted countries meeting minimum proficiency standards in mathematics, to the point it never predicted any member of the test set was below minimum proficiency. Fortunately, the Gaussian Naive Bayes model performed admirably with both classes, with accuracy, precision, recall, and F1 all at or above 71 percent. Given the limited size of the data set, it seems likely that these scores would tend to improve with a larger test set to draw from. From the ROC curve, it is also clear that the naive bayes classifier model either matched or outperformed the support vector machine at all threshold levels.

On this basis, a mean difference analysis of feature importance was conducted, summarized in the following chart:

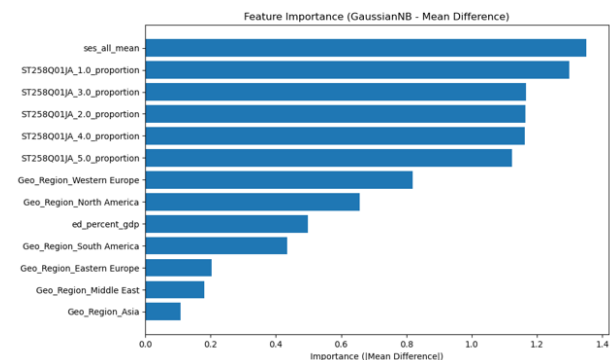


Figure 10: Mean Difference Feature Importance for Gaussian Naive Bayes Model of Minimum Math Proficiency

The most important feature in the model was the mean socio-economic status of the country, closely followed by the proportions of students experiencing various levels of food insecurity. A tertiary predictor was whether or not a student lived in the global west. Interestingly, the percent of GDP a nation spends on education, while still important, is dwarfed in significance (as measured by mean difference) by the previous three indicators. As such, we might conclude that, while education spending is an important predictor of math proficiency, ensuring stable access to the fundamental needs of living is of even greater import in supporting student success.

2.4 Aim 3 Evaluation and Results: Relationship between Math Proficiency and Food Security

We explored the impacts of food availability on average math performance level. Following PISA guidelines, we categorized an average math proficiency score below 420.07 as does not meet the minimal standards for proficiency. A score of 420.07 to below 482.38 meets the minimum standard, but not higher, and a score of 482.38 and above is considered above minimum standards. We also calculated the weighted proportion of students by country who, in the last 30 days, did not eat because there was not enough money to buy food, a) Never or almost never, b) About once a week, c) 2 to 3 times a week, d) 4 to 5 times a week, or e) Every day or almost every day. For the purposes of exploring the data, we considered missing food two or more times per week a significant impact on food availability, then calculated for each country what percentage of students met this criteria. When we investigated the univariate relationship of hunger and math proficiency, we found that the countries that on average scored below the minimum proficiency on math had a higher median number of students who did not eat two or more times per week when compared to countries where students on average met or exceeded the minimum math proficiency (Figure 11).

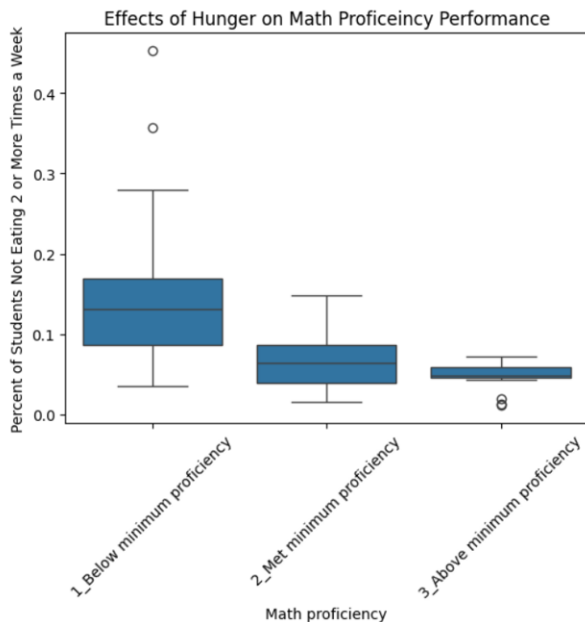


Figure 11: Percentage of students who did not eat two or more times per week by math proficiency level

When modeling this three-level outcome, we chose

random forest as it can accommodate non-binary classification problems. Additionally, the ensemble nature of the model means that it is likely to perform better than a decision tree, improving accuracy and predictions of the model. We also chose K-nearest neighbor (KNN) as we anticipated some grouping of countries by our features and KNN accommodates a 3-level outcome variable.

Random forest and KNN models require limited statistical assumptions. They do not require linearity, normality, or homoscedasticity. However, for random forest we assume that 1) each tree makes its own predictions independent of the other trees, 2) random samples of the data and features are used for each tree, 3) sufficient data are available, and 4) combining the predictions from the different trees improves the results. For KNN to work well we scaled our parameters so that we have meaningful distances between points.

For random forest, we completed hyperparameter tuning varying n-estimators (300, 400 and 500) and max-depth (2, 3, 5, and 8). We used balanced class weights, since we have an imbalance in our classes. Our optimal random forest model used n-estimators=300 and max depth = 2. When hyperparameter tuning our KNN model, we tried n-neighbors of 2, 3, 4 and 5. We also considered metrics of euclidean, max, manhattan, and minkowski. Our optimal KNN model used n-neighbors = 2 and metric = euclidean.

We faced some challenges with fitting the model. We first tried to use categorical SES and GDP measures based on quartiles and World Bank cut-offs, respectively. We found that relationships we know exist between SES, GDP and math scores were not reflected in the models. When we instead ran the models with continuous normalized SES and GDP, we saw that these variables were related. We also dropped region, as it did not provide meaningful information for either model. Additionally, the KNN model performed better with fewer predictors. We were able to determine that the only important variables were SES, GDP and food security and limited the KNN models to these features.

Model	Actual	Pred: < min prof	Pred: min prof	Pred: > min prof	precision	recall	f1-score
Random Forest < min proficiency	8	1	1	0	0.800000	0.888889	0.842105
Random Forest min proficiency	2	1	1	3	0.500000	0.166667	0.250000
Random Forest > min proficiency	0	0	0	5	0.625000	1.000000	0.769231
KNN < min proficiency	8	0	0	1	0.727273	0.888889	0.800000
KNN min proficiency	2	2	2	2	0.666667	0.333333	0.444444
KNN > min proficiency	1	1	1	3	0.500000	0.600000	0.545455

Table 5: Evaluation metrics: Accuracy, Precision, Recall, F1-score

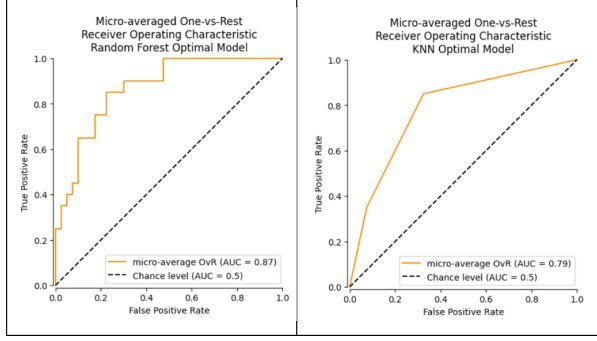


Figure 12: Micro-Averaged One-vs-Rest ROC-AUC for Math Proficiency Level

We found that in general the random forest model performed better than the KNN model in classifying the countries with average student performance below the minimum proficiency (precision=0.80, recall=0.89 and f1-score=0.84 compared to precision=0.72, recall=0.89 and f1-score=0.80) and above the minimum proficiency (precision=0.65, recall=1.00 and f1-score=0.77 compared to precision=0.50, recall=0.60 and f1-score=0.54) levels (Table 5 and Figure 12). Both models did poorly classifying the countries that met minimum proficiency on average, but did not score higher than that on average. Although KNN did slightly better than random forest, neither model performed better than chance for this middle group of countries.

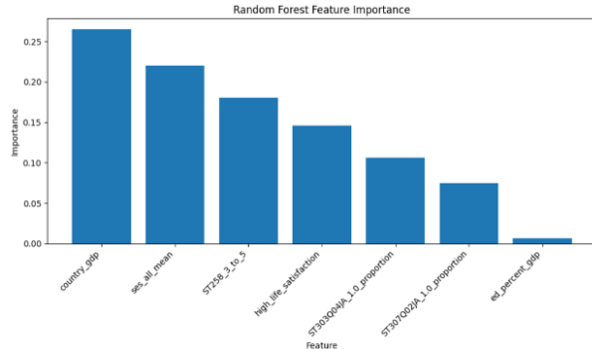


Figure 13: Gini Feature Importance for Random Forest Model of Average Math Proficiency Level

Our random forest model supported this finding. Our partial dependency plots (data not shown) among the countries with average student proficiency below the minimum standards found higher levels of food insecurity related to higher levels of students not meeting the minimum standards. As expected, the proportion of students not meeting the minimum standard decreases as average SES and country-level GDP increase. We also looked at variable importance

and found that the top variables were country-level GDP, average SES and food insecurity (Figure 13).

3 Conclusion and Future Work

Using linear regression, we were able to establish a predictive relationship between a student's reading ability and performance on measures of math proficiency. When considering behavioral variables as well, a student's subjective love of learning improved the model for predicting math proficiency. Based on the evaluation metrics RMSE and adjusted R^2 , this two feature model was ultimately the model that we thought did the best at predicting math proficiency using a linear relationship.

When testing binary classifier models, we implemented Gaussian Naive Bayes and SVM models to determine the ability of the models to predict average student math scores in a country based on several factors, including average socio-economic status, percent of GDP spent on education, and proportions of students at various levels of food insecurity. Overall, the Naive Bayes classifier significantly outperformed SVM, and analysis of the contribution of each feature to the model indicated that the most predictive features were average socio-economic status and level of food insecurity.

Further exploration of the potential relationship between food insecurity and math proficiency was undertaken using a Random Forest ensemble model and KNN. The response variable was partitioned into three categories (below minimum proficiency, met minimum proficiency, and exceeded minimum proficiency), and total national GDP was added as a feature. The Random Forest model performed best overall of the two, though both models struggled to predict countries that fell in the middle category better than chance would. Interestingly, these models rated food security as of tertiary importance as a predictor, behind both average socio-economic status and national GDP.

Our models allowed us to establish several key relationships within the educational framework. First, as expected, high and low performance in different academic domains tend to be highly correlated (as demonstrated by our linear regression model). Second, some social factors, such as average socio-economic status, national GDP, and level of food insecurity, tend to be more predictive of performance on measures of math proficiency than geographic region or those behavioral variables included in our models. Interestingly, we also discovered that the percent of GDP a country spends on education, while still sig-

nificant to the models in which it was included, was not one of the features most important to the performance of the models. This fact would seem to warrant further investigation, preferably with more granular data breaking down how countries spend on education relative to their overall monetary investment. Such investigation could potentially lead to fruitful avenues of research into more efficient ways to allocate scarce resources within the education system to facilitate more optimal outcomes for students.

Despite having access to massive, comprehensive datasets of student-level performance data (in conjunction with student, teacher, and school level responses to social-emotional questionnaires), the opacity of the methods by which these data points were converted into overall proficiency scores forced us to rely on country-level aggregated data for student academic performance. Having recently learned about the potential for unsupervised learning models to find and create categories within uncategorized data, we believe that potential future work with this data could include a preliminary phase in which unsupervised models are used to construct student-level clusters of proficiency on measures of math, reading, and science. These generated clusters could then be used to perform much more granular analysis of the trends we uncovered at the student, rather than country, level. In conjunction with additional data on specific education spending by countries, the potential understanding of the various factors which predict student success are immense.

4 References

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